INCH-POUND

MIL-P-50206 (AR)
w/ AMENDMENT 5
27 June 2008

SUPERSEDING
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w/ AMENDMENT 4
22 January 1998

MILITARY SPECIFICATION

PROPELLANT M9, FLAKE
(FOR USE IN 40MM CARTRIDGES)

This specification is approved for use by the Department of the Army and is available for use by all Departments and Agencies of the Department of Defense

1. SCOPE

1.1 This specification covers M9 Flake Propellant for use in 40MM cartridges (see 6.3).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

STANDARDS

MILITARY

MIL-STD-l05	Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-286	Propellant, Solid, Sampling, Examinations and Testing
MIL-STD-652	Propellant, Solid for Cannons, Requirements and Packing
MIL-STD-1168	Lot Numbering of Ammunition

DRAWINGS

ARMY

C-122-2-1B	Head Assembly, Edgewood Arsenal
D8844609	Case, Cartridge, Loading Assembly, XM118, Picatinny Arsenal
C9207988	Case, Cartridge, Loading Assembly, M195, Picatinny Arsenal

Comments, suggestions, or questions on this document should be addressed to: Commander, US Army ARDEC, Attn: AMSRD-AAR-QES-E, Picatinny, New Jersey 07806-5000 or emailed to ardecstdzn@conus.army.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST online database at http://assist.daps.dla.mil.

FSC 1376

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C8864622 Separation Charge Assembly for Fuze, M536 40MM Cartridge M397

9922230 Case, Cartridge, Practice M212

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Un1ess otherwise indicated, the issue in effect on date of invitation for bids shall apply.

CODE OF FEDERAL REGULATIONS

Title 49 Transportation, Parts 49 CFR 171-179

The Interstate Commerce Commission Rules are now a part of the Code of Federal Regulations, available from the Superintendent of Documents, U. S. Government Printing office, Washington, D.C., 20402. Order. for the above publication should cite "49 CFR 171-179, (latest revision)."

3. REQUIREMENTS

3.1 Materials. The propellant shall be M9, Flake, conforming to the requirements of Standard MIL-STD-652, and drawings (dwg) listed in Table I, except that the graphite content shall be 0.4 percent, maximum (max.).

TABLE I

Cartridge Designation	Drawing
M118	D8844609
Head Assembly	C122-2-18
M195	C9207988
M397	C8864622
M212	9322230

3.2 Form. The propellant shall be in the form of solid discs when examined as specified in 4.3.2.

3.3 Physical properties. The physica1 properties shall be as specified in Table II when tested as specified in 4.3.1 and 4.3.2.

TABLE II

Diameter, inches (advisory)	0.035
Thickness, inches (advisory)	0.00325
Bulk density lbs/cu.ft., min.	30.0

3.4 Closed Bomb, Relative Quickness and Relative Force. The M9 propellant shall have a relative quickness and a relative force of 100 plus or minus 3 percent when tested as specified in 4.3.3.

3.5 Workmanship. The best commercial practices shall be used in the formulation of propellant furnished under this specification, and all other applicable documents. The propellant and its standard

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ingredients shall be protected from the action of direct sunlight and acid fumes. No reground propellant or nitrocellulose shall be used in the manufacture of propellant procured under this specification unless otherwise specified.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and serv1ces conform to prescribed requirements. Reference shall be made to Standard MIL-STD-l09 in order to define terms used herein.

4.1.1 Submission of product. At the time the completed lot of product is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and app11cable to the product being submitted:

a. A statement that the lot complies with all of the qua1ity assurance provisions specified within this specification.

b. Number of batches of product inspected, (shall be made available upon request by the contracting officer).

c. Results obtained, by defect code, for all inspections performed, (shall be made available upon request by the contracting officer).

d. Drawing, specification number and date, together with an identification and date of change.

e. Certificate. of conformance an all material purchased by the contractor when such material is controlled by Government or commercial specification. referenced in any of the contractual documents.

f. Number of pounds in the lot.

g. Date submitted.

h. Eight copies of propellant. description sheet, AMC Form 1047) (00 Form 1204) containing the applicable data (see 6.4).

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent’s authority to bind his principal. Substantiation of the agent’s authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 Inspection provisions

4.2.1 Lot formation. The term "lot," as used throughout this specification, refers to an inspection lot, which is defined as an essentially homogeneous quantity of propellant from which the representative sample is drawn and inspected to determine conformance of the lot with applicable requirements. The

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sample selected sha11 represent only that quantity of propellant from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. A lot shall consist of one or more batches of M9 propellant which has been subjected to the same unit chemical or physical process intended to make the final product homogeneous. Unless otherwise approved by the contracting officer, the inspection lot size of propellant deliverable under the contract shall be not less than the smallest week1y estimate of quantities contractually scheduled for production during the contract period nor more than the largest quantity contractually scheduled for delivery during any month of the contract period. The classification of defects shall be in accordance with Standard MIL-STD-105. The criteria and procedures for the assignment of lot numbers shall be in accordance with Standard MIL-STD-ll68.

4.2.2 Testing

4.2.2.1 Sampling. The sampling shall be in accordance with Standard MIL-STD-652, except that a five-pound sample shall be used for chemical and physical tests. If the sample fails to meet the requirements of paragraph 3, the lot shall be rejected.

4.3 Test methods and procedures

4.3.1 Bulk density, Major defect, Code No. 01001. The bulk density shall be determined in accordance with Method 502.2 or 502.1 of Standard MIL-STD-286.

4.3.2 Form and dimensions, Major defect, Code No. 02001. One hundred flakes shall be examined under a microscope for form and dimension.

4.3.3 Closed bomb test, Major defect, Code No. 03001. The propellant shall be tested in accordance with Method 801.1 of Standard MIL-STD-286 (see 6.4 and 6.5).

4.3.4 Determination of graphite, Minor defect, Code No. 04001. The graphite content shall be determined as specified in MIL-STD-286, Method 309.1.

5. PREPARATION FOR DELIVERY

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DOD or in-house personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Process. - Details of the manufacturing process and the equipment used by the contractor will be submitted to the procuring activity, in writing, prior to commencement of manufacture.

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6.2 Inspection code numbers. The five-digit code numbers, assigned to the inspections herein, are to facilitate future data collection and analysis by the Government.

6.3 Intended use. It is intended that the item of this specification will be used in the following case cartridges: M118, Ml95, M397, and the head assembly.

6.4 Submission of AMC Form l047. Information contained in the propellant description sheet, AMC Form 1047, need not be resubmitted in the form of a certificate, provided the provisions regarding certifying of responsible agents is adhered to (see 4.1.1). The Form 1047 shall be forwarded to: Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND2 and SMUPA-DE2, Dover, New Jersey, 07801.

6.5 Closed bomb test. Lot No. RAD06K-G66695 should be used as a standard in the closed bomb test.

6.6 Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Custodian:
Arrmy-AR

Preparing activity:
Army-AR
(Project Number 1376-2008-009)

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